Software Test Plan (STP) - Canteen Pre-Order System

Project: Canteen Pre-Order System
Version: 1.0
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1. Introduction

Purpose: This document defines the test plan for the Canteen Pre-Order System v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers student registration and login, menu browsing, cart management, pre-order placement, payment-status handling, order tracking, order history, cancellation, staff menu management, food availability, and order processing. Physical food preparation and activities outside the software are excluded.

References: Canteen Pre-Order System SRS v1.0, Software Architecture and Design Specification v1.0, Requirements Traceability Matrix (RTM).

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design Specification), STP (Software Test Plan), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing).

2. Test Items

- Student authentication and user management module
- Menu browsing and search module
- Cart and checkout module
- Order placement and order-history module
- Payment-status module
- Order tracking and cancellation module
- Staff authentication and authorization module
- Menu management and food-availability module
- Staff order queue and status management module

3. Features to be Tested

Features mapped to SRS requirement IDs:
- CPS-F-001: User registration
- CPS-F-002: User login
- CPS-F-003: Menu display
- CPS-F-005: Cart management
- CPS-F-007: Order placement
- CPS-F-009: Payment-status handling
- CPS-F-010: Order tracking
- CPS-F-012: Order cancellation
- CPS-F-014: Menu management
- CPS-F-016: Order status update
- CPS-NF-001: Response time
- CPS-NF-008: Data integrity
- CPS-SR-003: Staff authorization

4. Features Not to be Tested

- Internal implementation of external payment providers; only the Canteen Pre-Order System's integration behaviour is tested.
- Physical food preparation, cooking, packaging, and other activities outside the software.
- Production payment infrastructure when a sandbox or simulated payment environment is used.
- Browser and operating-system internals.

5. Test Approach / Strategy

Levels:
- Unit tests (module-level)
- Integration tests (frontend ↔ backend ↔ database)
- System tests (end-to-end student and staff functionality)
- Acceptance tests (UAT)

Types:
- Functional testing (core features)
- Regression testing
- Performance testing (response time, load)
- Usability testing (UI clarity and responsive behaviour)
- Security testing (authentication, authorization, input validation)
- Compatibility testing (supported browsers)

Entry Criteria: Stable build delivered, test data available, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, and all high-priority acceptance criteria satisfied.

5.1 Security Validation

- Validate password handling (hashing and no plaintext logging)
- Authentication and session validation
- Role-based authorization for staff functions
- Server-side validation of prices, quantities, item IDs, availability, and order totals
- HTTPS verification in deployed environments
- Input validation and authorization-bypass testing

6. Test Environment

Hardware: Standard laptop, desktop, and mobile devices; no specialized hardware is required for core testing.

Software: Canteen Pre-Order System v1.0, Node.js/Express backend, MongoDB test database, and supported web browsers.

Tools: Postman (API testing), optional Jest/React Testing Library (test automation), optional JMeter (performance testing), Git/GitHub (source control).

Test Data: Dummy student accounts, staff accounts, menu items, cart contents, orders, and payment-status records.

7. Test Schedule

Milestones:
- Test case design: 05-Oct-2026 to 10-Oct-2026
- Environment setup: 11-Oct-2026 to 13-Oct-2026
- Sprint 1 testing: 19-Oct-2026 to 23-Oct-2026
- Sprint 2 testing and regression: 26-Oct-2026 to 30-Oct-2026

- Final defect verification: 31-Oct-2026 to 01-Nov-2026
- UAT: 02-Nov-2026 to 10-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Rohini Vishu	Prepare plan, coordinate test execution
Test Engineer	Sumeet Shiggavi	Design & execute test cases, log defects
Developer	Sindhu K S	Support defect fixes and triage
Product Owner	Pooja L R	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the development team
Test environment downtime	Maintain a backup test environment and database backup
Dependency on external payment-service sandbox	Use a stable sandbox/test stub and verify integration behaviour separately

11. Assumptions & Dependencies

- Canteen Pre-Order System test environment will be stable and available.
- Test data for student accounts, staff accounts, menu items, orders, and payment-status records will be provided before execution.
- Required backend services, database, and test tools will be available during testing.

12. Suspension & Resumption Criteria

Suspend testing if:
- Test environment is unavailable for more than 4 hours.
- The build is too unstable and blocks more than 30% of planned test cases.

Resume testing if:
- Blocking defects are resolved.
- The test environment is stabilized.
- A stable build is delivered for retesting.

13. Test Case Management & Traceability

RTM ensures mapping of Canteen Pre-Order System SRS requirements to test cases.

Example:
- CPS-F-001 (User registration) → TC-AUTH-01
- CPS-F-002 (User login) → TC-AUTH-02
- CPS-F-007 (Order placement) → TC-ORD-01
- CPS-F-008 (Order confirmation) → TC-ORD-02
- CPS-F-014 (Menu management) → TC-STAFF-02
- CPS-NF-001 (Response time) → TC-PERF-01
- CPS-NF-008 (Order data integrity) → TC-DATA-01
- CPS-SR-003 (Staff authorization) → TC-SEC-03

14. Test Metrics & Reporting

Metrics collected:
- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:
- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Rohini Vishu	11-10-2026
Dev Lead	Sindhu K S	11-10-2026
Product Owner	Pooja L R	11-10-2026